Fiscal Year:	FY 2023	**Task Last Updated:**	FY 09/29/2022
PI Name:	Steen, Paul H Ph.D.		
Project Title:	Capillary Flows and Draining in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	Ithaca	**State:**	NY
Zip Code:	14853-5201	**Congressional District:**	23
Comments:	PI is deceased as of September 2020		
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-D: Use of the NASA Physical Sciences Informatics System – Appendix D
Start Date:	01/31/2019	**End Date:**	04/30/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Hatch, Tyler	**Contact Phone:**	216.433.5073
Contact Email:	tyler.r.hatch@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 4/30/2022 (previously extended to 12/31/2021) with Olivier Desjardins as PI, overseeing Josh McCraney's research after Paul Steen passed away (Ed., 12/2/21)		
Key Personnel Changes/Previous PI:	NOTE (Ed., November 2020): Principal Investigator Dr. Paul Steen passed away in September 2020; Dr. Olivier Desjardins will now oversee Joshua McCraney's research on this project. NOTE (Ed., Mar 2019): This work is result of proposal by graduate student Joshua McCraney, Ph.D. candidate at Cornell University, who will be performing the work with advisor PI Dr. Paul Steen.		
COI Name (Institution):	McCraney, Joshua Ph.D. (Graduate Student (PostDoc May 2021) PI: Cornell University)		
Grant/Contract No.:	80NSSC19K0406		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Capillary Flow Experiments (CFE) for Interior Corner Flows (ICF) were conducted aboard the International Space Station (ISS) and rely solely on video for high-quality quantitative data. We propose to digitally measure the still video images of the time-dependent interface height. Since container geometry is known, the meniscus heights can be used to determine volumetric drain rates. This data would then serve as both a benchmark for current theory on drain rates and a guide for extending capillary flow models to a diverse set of geometries and flow conditions. We also propose to utilize and extend a numerical tool to simulate 3D transient corner flows with and without topological change. To pursue any discrepancy between data and numerics, supportive ground-based experiments at micro-scales and low-gravity experiments conducted in the Dryden Drop Tower at Portland State University are proposed. Understanding the dynamics of capillary flows and responses of drain flows to various geometries offers design advantages for spacecraft waste-management systems, fuel control, hydration containment, and cryogenic flows. Capillary flows and drain problems are relevant to a broad range of two-phase flow technologies in low gravity, including 3D vane networks for positioning of fuels, an important problem concerning propellant liquid management aboard spacecraft.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The capillary drain flows see immediate terrestrial application for biofluids during lab-on-chip processing. For low-gravity environments, capillary-dominated fluid behavior is common over large length scales in the near-weightless environment aboard spacecraft. In part due to both the expense and limited opportunities for designers to develop significant intuition of general multiphase phenomena in such environments, developing robust fluid systems for spacecraft fuels management, thermal control, and water processing for life support remains challenging. For the specific example of liquid fuels aboard orbiting spacecraft, capillary draining can serve as a limit to the life of the spacecraft if and when the residual fuel in the tank is or becomes inaccessible. It is essential to establish the maximum capillary flow rate at which a container in a micro-gravity environment can be drained to increase satellite predictability and lifespan (when fuel is empty, the satellite is essentially expired).
Task Progress:	Compiled from the final technical report submitted by Dr. McCraney (Ed., September 2022). The grant proposal set out to experimentally benchmark low gravity drain flows and to extend modeling and computational capabilities to inertial capillary flow models. This was to be achieved in four tasks: 1) mine the Interior Corner Flow (ICF) database in the NASA Physical Sciences Informatics (PSI) video archive for draining flows and compare to current corner flow theory; 2) benchmark and extend a numerical tool to model 3D transient corner drain flows; 3) resolve discrepancies amongst mined data and models through computational solutions and ground experiments; 4) exploit inertial ICF data identified in task 1 to guide developing theory in inertial-capillarity. All four tasks were achieved and published in multiple papers. [Ed. Note: see cumulative bibliography.] New technology was designed during the duration of the grant, and explicitly aided in the publication of several papers. This has been reported to NASA, and a summary follows: a MATLAB script to automate image tracking was designed. Specifically, the technology automates identifying the location of image lines. While the implementation was demonstrated on fluid systems where the liquid-gas interface shifted, this technology could be used in multiple applications beyond the intended scope.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Articles in Peer-reviewed Journals	McCraney J, Weislogel M, Steen P. "Capillary flow experiments conducted aboard the International Space Station: Experiments and simulations." Microgravity Sci Technol. 2022 Jul 18;34:63. https://doi.org/10.1007/s12217-022-09988-7 , Jul-2022
Articles in Peer-reviewed Journals	McCraney J, Kern V, Bostwick JB, Daniel S, Steen PH. "Oscillations of drops with mobile contact lines on the International Space Station: Elucidation of terrestrial inertial droplet spreading." Phys Rev Lett. 2022 Aug 19;129(8):084501-7. https://doi.org/10.1103/PhysRevLett.129.084501 , Aug-2022
Articles in Peer-reviewed Journals	Ludwicki JM, Kern VR, McCraney J, Bostwick JB, Daniel S, Steen PH. "Is contact-line mobility a material parameter?" npjMicrogravity. 2022 Feb 21;8(6): https://doi.org/10.1038/s41526-022-00190-y , Feb-2022
Articles in Peer-reviewed Journals	McCraney J, Bostwick J, Weislogel M, Steen P. "Bubble migration in containers with interior corners under microgravity conditions." Exp Fluids. 23 Jul 27;64:1402. https://doi.org/10.1007/s00348-023-03677-w , Jul-2023
Articles in Peer-reviewed Journals	McCraney J, Ludwicki J, Bostwick J, Daniel S, Steen P. "Coalescence-induced droplet spreading: Experiments aboard the International Space Station." Phys. Fluids. 2022 Dec 13;34:122110. https://doi.org/10.1063/5.0125279 , Dec-2022